

RNA-seq Analysis Report

RNA-seq: Romidepsin+IFN vs Untreated in BRAF-WT and BRAF-MUT Melanoma
(SRP414072)

OmicsDesk

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1 Executive Summary

Romidepsin combined with IFN treatment drives substantial transcriptional remodeling in both BRAF-mutant and BRAF-wild-type melanoma cell lines, though the magnitude of response differs markedly between genotypes. Across two pairwise comparisons — treated versus untreated within each BRAF background — the analysis identifies 44 differentially expressed genes in BRAF-mutant cells (33 up, 11 down) and 142 in BRAF-wild-type cells (109 up, 33 down), suggesting that BRAF-wild-type tumors mount a broader transcriptional response to this combination therapy. The most striking upregulated signals in both comparisons center on interferon-stimulated genes. IFI27, IFI44L, ISG15, OAS2, and CMPK2 are strongly induced in BRAF-mutant cells, while IFI27 and IFI44L are among the highest-confidence hits in BRAF-wild-type cells as well, confirming robust activation of the type I interferon pathway across both genotypes. Pathway enrichment analysis reinforces this picture and simultaneously reveals a consistent and highly significant suppression of cell cycle programs — particularly chromosome segregation, DNA replication, and mitotic spindle checkpoint pathways — in both comparisons, indicating that the treatment effectively arrests proliferation regardless of BRAF status. Transcription factor activity analysis further supports these findings: STAT2, IRF9, and STAT1 emerge as the most strongly activated regulators in both contexts, consistent with canonical interferon signaling, while the E2F family transcription factors — key drivers of cell cycle entry — are prominently repressed. Together, these results point to a dual mechanism of action: immune pathway activation coupled with cell cycle shutdown.

2 Cross-Analysis Observations

This section is generated **directly from the data** — PCA scores, differential analysis counts, pathway enrichment results, and (where available) regulator activity — by a rule-based synthesis layer (no machine-generated prose). It surfaces the cross-analysis story that an analyst would normally write by hand after looking at the figures together.

Auto-interpretation file not found — re-run the `auto_interpretation` Snakemake rule.

3 Methods

3.1 Sequencing Quality Control

Raw reads were processed with **fastp** (v0.23+) to remove adapter sequences and low-quality bases ($Q < 20$). Reads shorter than 36 bp after trimming were discarded. Per-sample reports were aggregated with **MultiQC** (v1.20+) into a single HTML QC report.

3.2 Alignment

Trimmed reads were aligned to the **GRCh38 (Ensembl 111)** reference genome using **subjunc** (Subread v2.0+) with splice-aware alignment. Gene-level read counts were quantified using **featureCounts** (Subread v2.0+) against Ensembl annotation (release matched to GRCh38 (Ensembl 111)).

3.3 Differential Expression

Differential expression analysis was performed using **DESeq2** (v1.40+). The Wald test was applied for all pairwise comparisons. Genes with fewer than 10 total reads across all samples were excluded (independent filtering). Significance threshold: adjusted p-value (Benjamini-Hochberg) < 0.05 , $|\log_2 \text{fold-change}| > 1$.

3.4 Pathway Enrichment

Gene Set Enrichment Analysis (GSEA) was performed using **clusterProfiler** (v4+) and **ReactomePA** against three curated databases:

- Gene Ontology Biological Process (GO:BP)
- KEGG Pathways
- Reactome Pathways

Enrichment significance threshold: $\text{FDR} < 0.05$. Top 15 terms per comparison and database are reported.

3.5 Transcription Factor Activity

Transcription factor activity was inferred using **decoupleR** (v2+) with the **DoRothEA** regulon database (confidence levels A, B, C; **doRothEA** R package) via the normalised weighted mean (NWM) method. Wald t-statistics from DESeq2 were used as the gene-level activity input. Top 20 TFs ranked by absolute NWM score are reported per contrast.

4 Quality Control

4.1 Alignment Statistics

Table 1: Alignment summary. Reads: after fastp trimming. Aligned: uniquely mapped reads.

Sample	Raw reads	Reads after QC	Q30 rate (%)	Uniquely mapped (%)	Multi-mapped (%)
BRAFWT_untreated_1	4247387	4133805	93.4	60.2%	0.0%
BRAFWT_untreated_2	4849354	4677323	93.2	60.8%	0.0%
BRAFWT_untreated_3	4118774	3970087	93.2	66.4%	0.0%
BRAFWT_treated_1	5125130	4931683	93.0	65.9%	0.0%
BRAFWT_treated_2	4434338	4251283	94.0	36.2%	0.0%
BRAFWT_treated_3	5435819	5221707	93.2	55.3%	0.0%
BRAFMUT_untreated_1	2769768	2667435	93.9	53.8%	0.0%
BRAFMUT_untreated_2	4151408	4012680	93.5	56.1%	0.0%
BRAFMUT_untreated_3	5390500	5206626	93.7	48.3%	0.0%
BRAFMUT_treated_1	5260961	5154190	94.0	39.9%	0.0%
BRAFMUT_treated_2	3739290	3684024	93.7	37.4%	0.0%
BRAFMUT_treated_3	5321604	5240596	93.7	51.0%	0.0%

MultiQC HTML report: /Volumes/Untitled/BioAnalytics/clients/geo_melanoma_20260317/output/qc/multiqc_report.htm

How to read this table: “Raw reads” is the size of the FASTQ file your samples came in as. “Reads after QC” is what survived our adapter-trimming and quality filtering (a small drop is normal — typically 1–3 %). “Q30 rate” is the share of bases the sequencer is highly confident about; values above 90 % indicate a clean run. “Uniquely mapped” is the share of reads that landed in exactly one place in the genome — anything above ~70 % is workable, above 85 % is excellent.

4.2 PCA — Sample Clustering

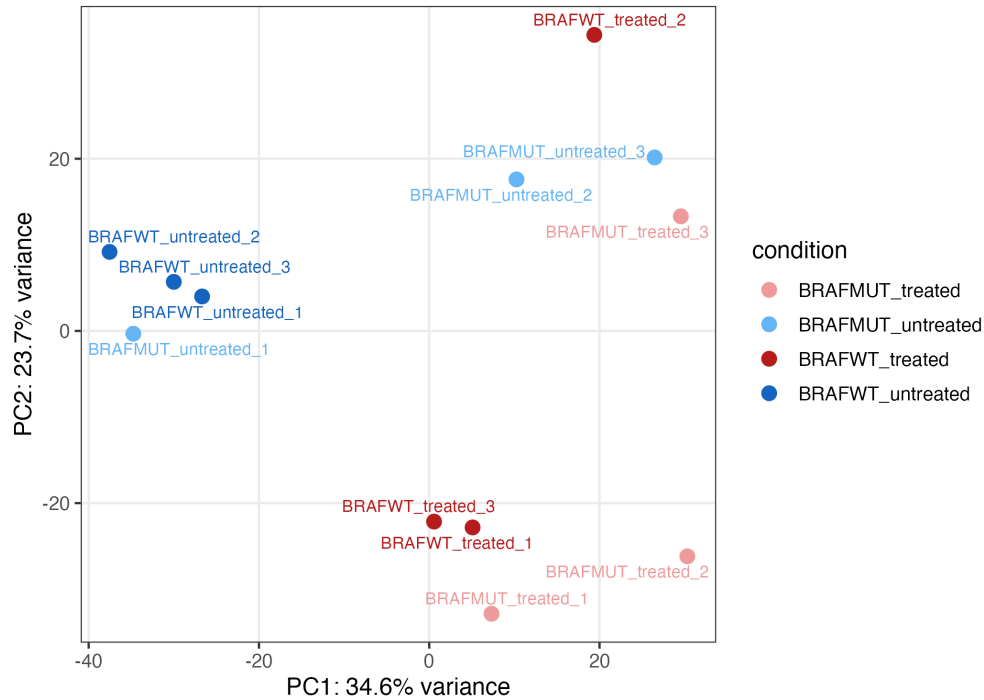


Figure 1: Principal component analysis of variance-stabilising transformed (VST) counts. Colours represent experimental conditions.

How to read this figure: Each dot is one sample. The two axes — PC1 and PC2 — are the two most influential summaries of the gene-expression differences between samples. If samples from the same condition cluster together (and separate from other conditions), it means the biological effect is the dominant signal in the data. If samples mix instead, that’s a yellow flag worth investigating before trusting the differential expression results.

4.3 Batch Effect Assessment

We test every metadata column against the principal components to flag any non-biological variable that might be driving the structure of the data. A “strong” verdict means the variable explains 30 % of the variance on PC1 or PC2. If a strong batch source is **already in the design formula**, it has been modelled out of the differential analysis; if it is **not**, our recommendation is to either add it to the formula or apply ComBat-Seq before re-running.

Batch assessment not available.

5 Differential Expression

5.1 Results Summary

Table 2: DEG counts ($\text{padj} < 0.05$, $|\log_2\text{FC}| > 1$)

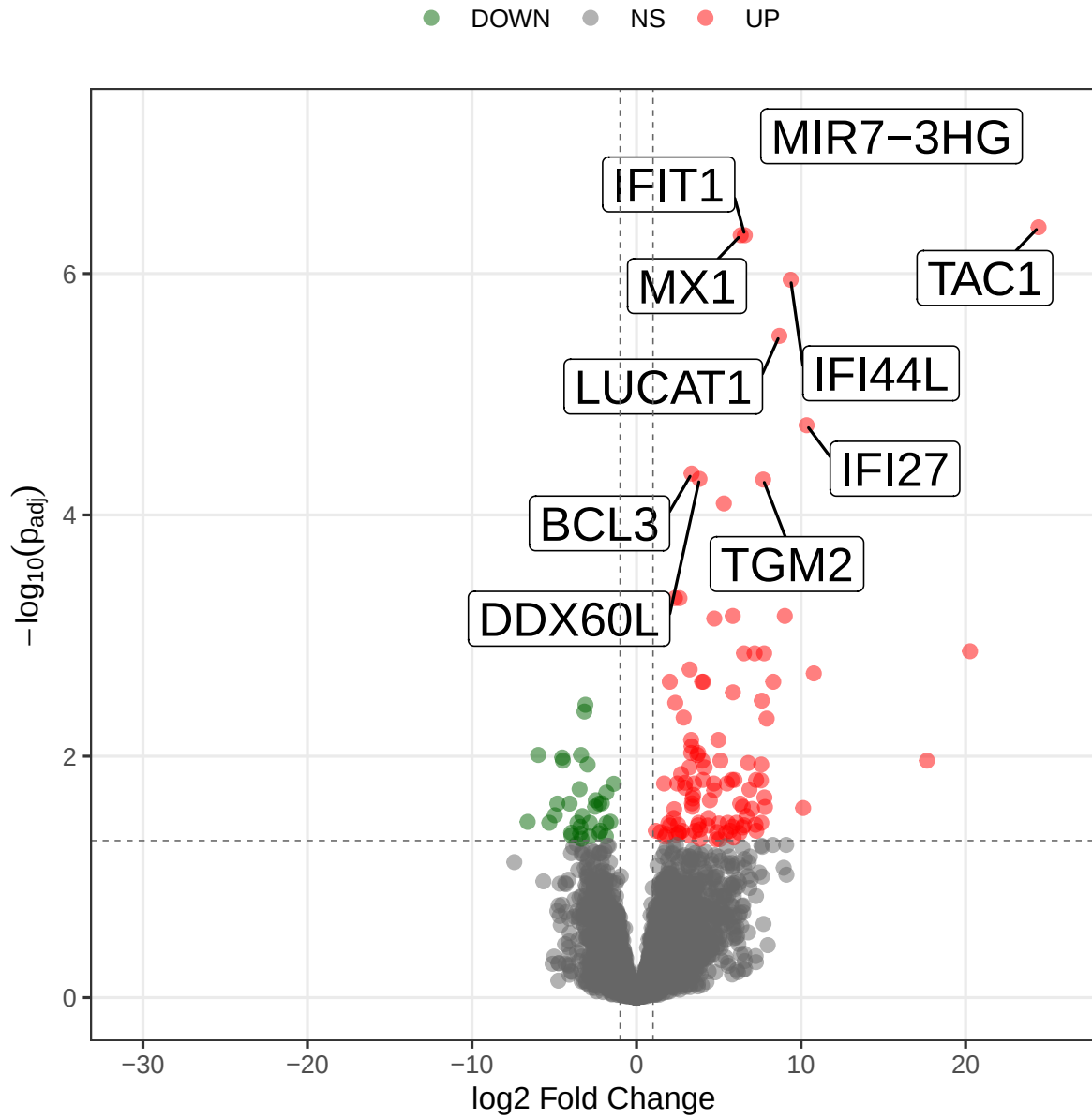
contrast	Total tested	Sig. UP	Sig. DOWN
BRAFMUT_treated_vs_BRAFMUT_untreated	20216	33	11
BRAFWT_treated_vs_BRAFWT_untreated	20216	109	33

How to read this table: Each row is one comparison (“contrast”). “Total tested” is the number of genes that had enough reads to be included in the statistical test. “Sig. UP” / “Sig. DOWN” count the genes that crossed both the significance threshold (adjusted p-value 0.05) and the effect-size threshold ($|\log_2\text{FC}| > 1$). A balanced UP/DOWN split is typical for broad perturbations; a strongly skewed split usually points to a directional biology (e.g. an inflammatory programme being switched on, or repressed).

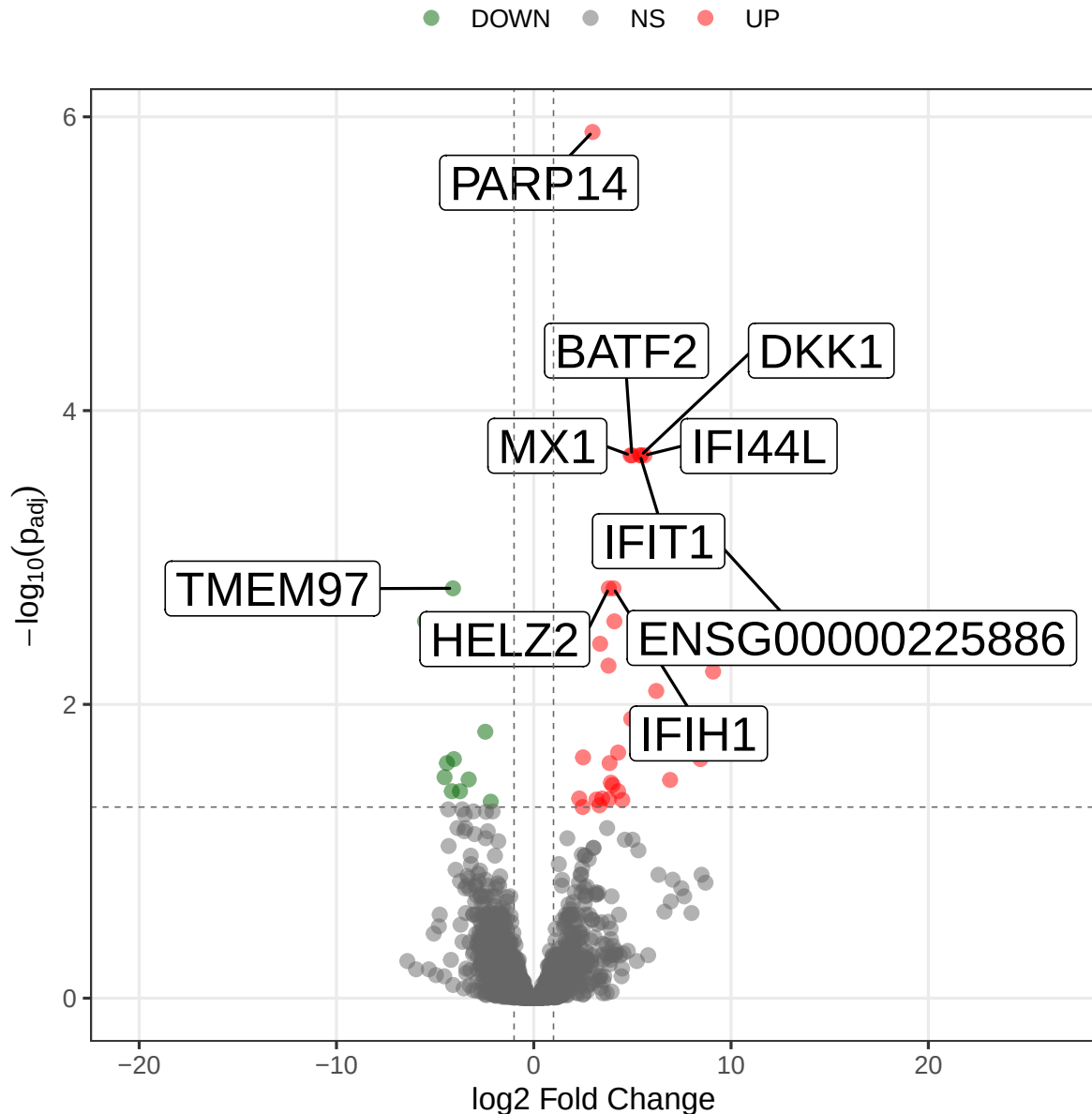
5.2 Volcano Plot(s)

How to read these plots: Each dot is one gene. The horizontal axis ($\log_2$ Fold Change) shows how much a gene moved up or down between the two conditions (positive = higher in the first group, negative = lower). The vertical axis ($-\log_{10} \text{padj}$) shows how confident we are: higher = stronger statistical evidence. Genes in the **top-right corner** are confidently up, **top-left** confidently down. Dashed lines mark the cutoffs used in this report. The 10 most significant genes per contrast are labelled.

BRAFWT_treated_vs_BRAFWT_untreated



BRAFMUT_treated_vs_BRAFMUT_untreated



5.3 Heatmap — Top 50 DEGs

How to read this figure: Each row is a gene, each column is a sample. Cell colour shows the gene's expression level relative to its average across all samples (red = above-average, blue = below-average — a “z-score”). The trees on the top and side group genes (and samples) that behave similarly. If columns from the same condition cluster together at the top dendrogram, the top DEGs are doing what we expect biologically.

Heatmap omitted: only 0 significant DEG(s) found across all contrasts (need 2).

5.4 DEG Overlap Across Contrasts

How to read this figure: The rows of dots at the bottom show which contrasts are being intersected (filled dots = “this contrast is in the group”). Each top bar is the size of that specific intersection — i.e. genes

significant in exactly those contrasts. The horizontal bars on the left show the total number of significant genes per contrast. UpSet plots scale better than Venn diagrams once you have more than three sets to compare.

UpSet plot omitted: at least 2 contrasts with significant DEGs are required.

5.5 Top Differentially Expressed Genes

These tables are the per-contrast leaderboard — the genes most worth eyeballing first. The full ranked DEG list (every contrast, every significant gene) is in the **results_workbook.xlsx** workbook delivered alongside this PDF.

Table 3: Top up-regulated DEGs per contrast (ordered by adjusted p-value).

Contrast	Gene	log2FC	Base Mean	padj
BRAFMUT_treated_vs_BRAFMUT_untreated	PARP14	2.98	266.6	1.30e-06
BRAFMUT_treated_vs_BRAFMUT_untreated	IFI44L	5.61	76.1	2.02e-04
BRAFMUT_treated_vs_BRAFMUT_untreated	BATF2	4.97	37.1	2.02e-04
BRAFMUT_treated_vs_BRAFMUT_untreated	DKK1	5.43	53.6	2.02e-04
BRAFMUT_treated_vs_BRAFMUT_untreated	IFIT1	5.39	116.9	2.02e-04
BRAFMUT_treated_vs_BRAFMUT_untreated	MX1	4.91	225.5	2.02e-04
BRAFMUT_treated_vs_BRAFMUT_untreated	ENSG00000225886	8.75	33.0	6.47e-04
BRAFMUT_treated_vs_BRAFMUT_untreated	IFIH1	4.04	41.0	1.62e-03
BRAFMUT_treated_vs_BRAFMUT_untreated	HELZ2	3.81	154.9	1.62e-03
BRAFMUT_treated_vs_BRAFMUT_untreated	STAT1	4.08	1217.2	2.72e-03
BRAFMUT_treated_vs_BRAFMUT_untreated	OAS2	5.44	48.9	2.72e-03
BRAFMUT_treated_vs_BRAFMUT_untreated	ISG15	5.46	300.0	3.42e-03
BRAFMUT_treated_vs_BRAFMUT_untreated	HERC5	3.36	143.7	3.87e-03
BRAFMUT_treated_vs_BRAFMUT_untreated	IRF7	3.79	31.9	5.45e-03
BRAFMUT_treated_vs_BRAFMUT_untreated	ANKRD1	9.09	378.6	5.98e-03
BRAFWT_treated_vs_BRAFWT_untreated	MIR7-3HG	22.54	35.6	1.00e-07
BRAFWT_treated_vs_BRAFWT_untreated	TAC1	24.43	105.5	4.00e-07
BRAFWT_treated_vs_BRAFWT_untreated	IFIT1	6.58	116.9	5.00e-07
BRAFWT_treated_vs_BRAFWT_untreated	MX1	6.33	225.5	5.00e-07
BRAFWT_treated_vs_BRAFWT_untreated	IFI44L	9.37	76.1	1.10e-06
BRAFWT_treated_vs_BRAFWT_untreated	LUCAT1	8.68	225.2	3.30e-06
BRAFWT_treated_vs_BRAFWT_untreated	IFI27	10.34	101.1	1.81e-05
BRAFWT_treated_vs_BRAFWT_untreated	BCL3	3.35	33.0	4.54e-05
BRAFWT_treated_vs_BRAFWT_untreated	DDX60L	3.83	103.1	5.03e-05
BRAFWT_treated_vs_BRAFWT_untreated	TGM2	7.70	49.8	5.09e-05
BRAFWT_treated_vs_BRAFWT_untreated	CKB	5.31	115.8	8.03e-05
BRAFWT_treated_vs_BRAFWT_untreated	BIRC3	7.65	138.2	3.07e-04
BRAFWT_treated_vs_BRAFWT_untreated	PARP14	2.33	266.6	4.90e-04
BRAFWT_treated_vs_BRAFWT_untreated	ZNFX1	2.61	373.1	4.90e-04
BRAFWT_treated_vs_BRAFWT_untreated	ISG15	5.85	300.0	6.87e-04

Table 4: Top down-regulated DEGs per contrast (ordered by adjusted p-value).

Contrast	Gene	log2FC	Base Mean	padj
BRAFMUT_treated_vs_BRAFMUT_untreated	TMEM97	-4.09	64.3	0.00162
BRAFMUT_treated_vs_BRAFMUT_untreated	AURKB	-5.51	48.7	0.00272
BRAFMUT_treated_vs_BRAFMUT_untreated	NME4	-3.22	110.5	0.00309
BRAFMUT_treated_vs_BRAFMUT_untreated	RPL10P3	-2.46	78.4	0.01540

Table 4: Top down-regulated DEGs per contrast (ordered by adjusted p-value). *(continued)*

Contrast	Gene	log2FC	Base Mean	padj
BRAFMUT_treated_vs_BRAFMUT_untreated	CDCA8	-4.05	43.4	0.02360
BRAFMUT_treated_vs_BRAFMUT_untreated	MYBL2	-4.40	38.3	0.02510
BRAFMUT_treated_vs_BRAFMUT_untreated	DHFR	-4.52	25.9	0.03130
BRAFMUT_treated_vs_BRAFMUT_untreated	MCM4	-3.30	62.9	0.03240
BRAFMUT_treated_vs_BRAFMUT_untreated	PBK	-4.15	56.4	0.03900
BRAFMUT_treated_vs_BRAFMUT_untreated	GAS7	-3.74	307.4	0.03900
BRAFMUT_treated_vs_BRAFMUT_untreated	ENSG00000286259	-2.18	64.9	0.04600
BRAFWT_treated_vs_BRAFWT_untreated	GJB1	-3.11	123.4	0.00374
BRAFWT_treated_vs_BRAFWT_untreated	PSAT1	-3.17	56.5	0.00426
BRAFWT_treated_vs_BRAFWT_untreated	SCML4	-5.97	9.5	0.00977
BRAFWT_treated_vs_BRAFWT_untreated	ZNF689	-3.37	53.0	0.00977
BRAFWT_treated_vs_BRAFWT_untreated	ITPKB	-4.52	82.4	0.01020
BRAFWT_treated_vs_BRAFWT_untreated	PBK	-4.47	56.4	0.01090
BRAFWT_treated_vs_BRAFWT_untreated	TLNRD1	-2.97	67.8	0.01170
BRAFWT_treated_vs_BRAFWT_untreated	TOP2B	-1.39	128.2	0.01690
BRAFWT_treated_vs_BRAFWT_untreated	CENPF	-3.46	498.4	0.01870
BRAFWT_treated_vs_BRAFWT_untreated	MYO5A	-1.83	223.1	0.02000
BRAFWT_treated_vs_BRAFWT_untreated	GPM6B	-2.48	66.0	0.02320
BRAFWT_treated_vs_BRAFWT_untreated	FARP2	-2.11	55.6	0.02460
BRAFWT_treated_vs_BRAFWT_untreated	TRIM2	-2.27	157.0	0.02470
BRAFWT_treated_vs_BRAFWT_untreated	SAPCD2	-4.81	7.7	0.02470
BRAFWT_treated_vs_BRAFWT_untreated	BUB1B	-4.07	24.2	0.02470

6 Pathway Enrichment

How to read these dot plots: Each dot is a biological pathway (a curated set of genes that act together — e.g. “interferon signalling”). The horizontal axis is the **Normalised Enrichment Score (NES)**: positive = the pathway’s genes are coordinately *up* in the contrast, negative = coordinately *down*. Dot **colour** reinforces direction (red = activated, blue = suppressed). Dot **size** is how many genes from the pathway are driving the call — bigger dots are more robust hits. We screen three databases — GO Biological Process (broad biology), KEGG (canonical signalling/metabolism), Reactome (curated expert-vetted pathways) — because each catches different things. Top 15 per contrast at FDR < 0.05.

GO:BP plot omitted: no significant terms in any contrast.

KEGG plot omitted: no significant pathways in any contrast.

Reactome plot omitted: no significant pathways in any contrast.

7 Transcription Factor Activity

Transcription factor (TF) activity was inferred from the differential expression signatures using **decoupleR** (v2+) with the **DoRothEA** regulon database (confidence levels A, B, C). The normalised weighted mean (NWM) statistic was used to quantify TF activity per contrast, weighting target gene expression changes by the mode of regulation.

Table 5: Top 20 transcription factors by inferred activity score per contrast (DoRothEA NWM).

Contrast	TF	Score	p-value	Activity
BRAFMUT_treated_vs_BRAFMUT_untreated	STAT2	15.055	0.02	Activated
BRAFMUT_treated_vs_BRAFMUT_untreated	IRF9	9.167	0.02	Activated
BRAFMUT_treated_vs_BRAFMUT_untreated	STAT1	7.640	0.02	Activated
BRAFMUT_treated_vs_BRAFMUT_untreated	FOXO3	6.048	0.02	Activated
BRAFMUT_treated_vs_BRAFMUT_untreated	NFKB1	5.668	0.02	Activated
BRAFMUT_treated_vs_BRAFMUT_untreated	JUN	5.332	0.02	Activated
BRAFMUT_treated_vs_BRAFMUT_untreated	STAT3	5.267	0.02	Activated
BRAFMUT_treated_vs_BRAFMUT_untreated	IRF1	5.107	0.02	Activated
BRAFMUT_treated_vs_BRAFMUT_untreated	FOXL2	4.707	0.02	Activated
BRAFMUT_treated_vs_BRAFMUT_untreated	SMAD4	4.590	0.02	Activated
BRAFMUT_treated_vs_BRAFMUT_untreated	FOS	4.540	0.02	Activated
BRAFMUT_treated_vs_BRAFMUT_untreated	ATF4	4.453	0.02	Activated
BRAFMUT_treated_vs_BRAFMUT_untreated	RELA	4.425	0.02	Activated
BRAFMUT_treated_vs_BRAFMUT_untreated	ELK1	4.381	0.02	Activated
BRAFMUT_treated_vs_BRAFMUT_untreated	ATF2	4.283	0.02	Activated
BRAFMUT_treated_vs_BRAFMUT_untreated	LEF1	-4.556	0.02	Repressed
BRAFMUT_treated_vs_BRAFMUT_untreated	TFDP1	-5.781	0.02	Repressed
BRAFMUT_treated_vs_BRAFMUT_untreated	MYCN	-6.115	0.02	Repressed
BRAFMUT_treated_vs_BRAFMUT_untreated	E2F2	-6.574	0.02	Repressed
BRAFMUT_treated_vs_BRAFMUT_untreated	E2F4	-8.642	0.02	Repressed
BRAFWT_treated_vs_BRAFWT_untreated	STAT2	9.982	0.02	Activated
BRAFWT_treated_vs_BRAFWT_untreated	IRF9	8.722	0.02	Activated
BRAFWT_treated_vs_BRAFWT_untreated	STAT1	5.512	0.02	Activated
BRAFWT_treated_vs_BRAFWT_untreated	NFKB1	4.963	0.02	Activated
BRAFWT_treated_vs_BRAFWT_untreated	SMAD3	4.378	0.02	Activated
BRAFWT_treated_vs_BRAFWT_untreated	RELA	4.299	0.02	Activated
BRAFWT_treated_vs_BRAFWT_untreated	THAP11	4.116	0.88	Activated
BRAFWT_treated_vs_BRAFWT_untreated	CEBPB	3.789	0.02	Activated
BRAFWT_treated_vs_BRAFWT_untreated	HIF1A	3.634	0.02	Activated
BRAFWT_treated_vs_BRAFWT_untreated	SREBF2	-3.924	0.02	Repressed
BRAFWT_treated_vs_BRAFWT_untreated	ATF3	-4.276	0.02	Repressed
BRAFWT_treated_vs_BRAFWT_untreated	FOXO1	-4.451	0.02	Repressed
BRAFWT_treated_vs_BRAFWT_untreated	E2F3	-4.484	0.02	Repressed
BRAFWT_treated_vs_BRAFWT_untreated	LEF1	-4.701	0.02	Repressed
BRAFWT_treated_vs_BRAFWT_untreated	TFDP1	-4.758	0.02	Repressed
BRAFWT_treated_vs_BRAFWT_untreated	MYC	-5.498	0.54	Repressed
BRAFWT_treated_vs_BRAFWT_untreated	E2F2	-5.657	0.02	Repressed
BRAFWT_treated_vs_BRAFWT_untreated	MITF	-5.779	0.02	Repressed
BRAFWT_treated_vs_BRAFWT_untreated	E2F1	-8.055	0.04	Repressed
BRAFWT_treated_vs_BRAFWT_untreated	E2F4	-9.635	0.02	Repressed

How to read this table: Each row is one transcription factor (TF). The **Score** is decoupleR's normalised weighted mean — positive values mean the TF's known target genes moved up together (the TF appears

“activated” in the contrast), negative values mean its targets moved down (“repressed”). Larger |Score| = stronger evidence. We report the top 20 per contrast.

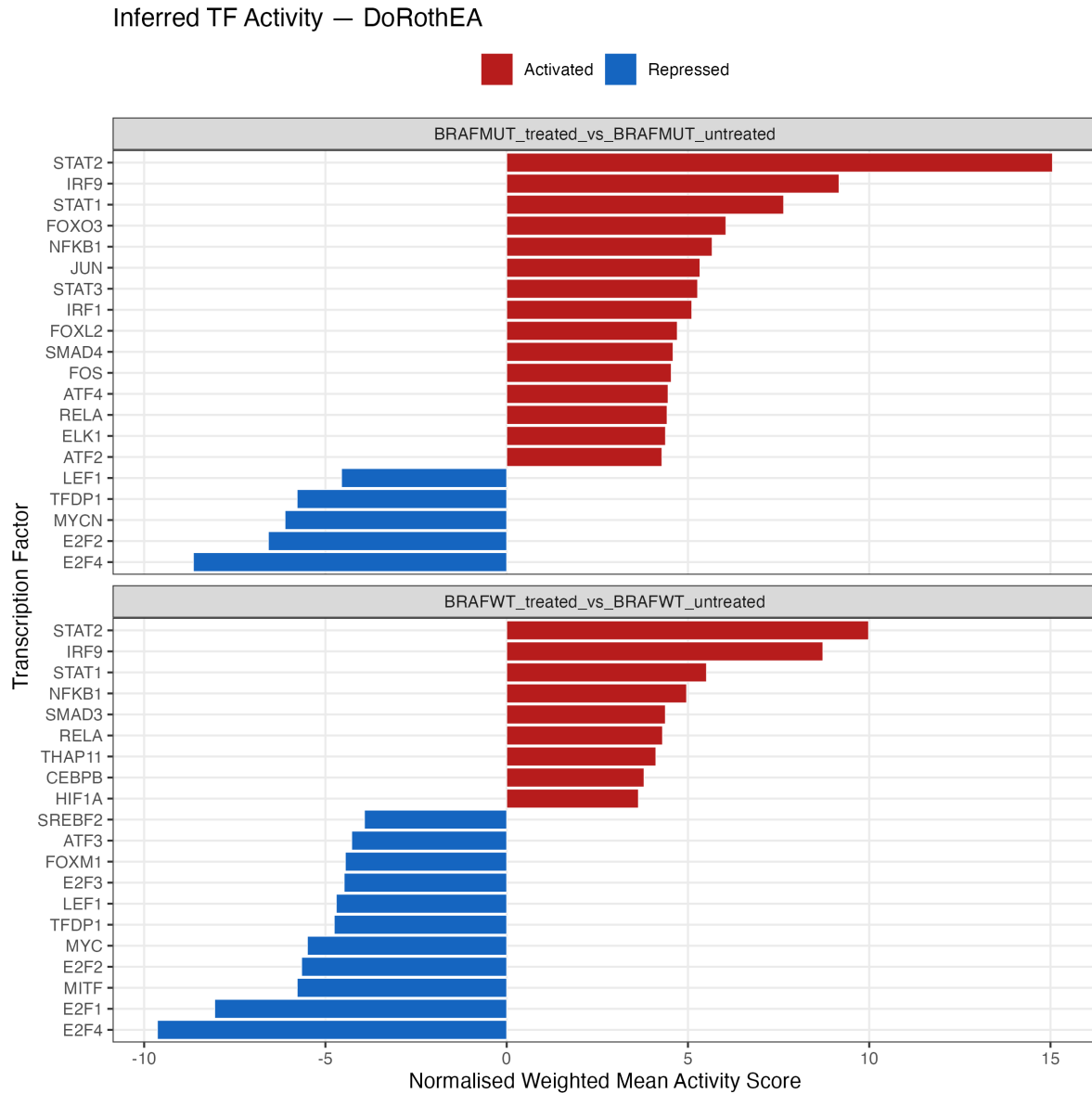


Figure 2: Top 20 transcription factors by inferred activity score per contrast. Red: activated TFs (positive NWM score); blue: repressed TFs (negative NWM score).



How to read this heatmap: Rows = TFs, columns = contrasts, cell colour = the same NWM score as above. Use it to spot TFs that flip direction across contrasts (mixed red and blue in the same row) versus TFs that move consistently. Only TFs with $|\text{score}| \geq 1$ in at least one contrast are shown to keep the figure readable.

8 Conclusions

Treatment with romidepsin combined with IFN drove a robust transcriptional response in both BRAF-mutant and BRAF-wild-type melanoma cells, characterised by two opposing programs: potent induction of interferon-stimulated genes and broad suppression of cell cycle machinery. Upregulated genes including IFI27, IFI44L, ISG15, OAS2, and CMPK2 aligned directly with the strongly enriched interferon alpha/beta signalling pathway, and this response was underpinned by coordinated activation of the STAT1–STAT2–IRF9 transcription factor axis in both genotypes. Simultaneously, downregulated genes spanning mitotic regulators—AURKB, PBK, MYBL2, CDCA8, BUB1B, and MCM4—explained the pervasive suppression of chromosome segregation, DNA replication, and mitotic spindle checkpoint pathways. Repression of E2F family members (E2F1, E2F2, E2F3, E2F4) and FOXM1 provides a clear mechanistic link to this anti-proliferative transcriptional shutdown. Notably, MITF and LEF1 were repressed in both contexts, suggesting concurrent attenuation of melanocyte identity and Wnt signalling. The BRAF-wild-type cells mounted a substantially larger response (109 versus 33 upregulated genes), indicating greater transcriptional sensitivity to the combination. To build on these findings, we recommend two targeted follow-up experiments: first, AURKB inhibitor co-treatment combined with romidepsin plus IFN to test whether further suppression of the residual mitotic program produces synergistic cell death; second, chromatin immunoprecipitation sequencing for STAT2 and IRF9 in BRAF-wild-type versus BRAF-mutant cells to determine whether the differential magnitude of the interferon response reflects genotype-specific differences in chromatin accessibility at interferon-stimulated gene promoters.

The complete DEG list (every contrast, every gene that passed the significance threshold) is delivered separately in **results_workbook.xlsx** — sheet “**Significant DEGs**” for filtered hits, “**All DEGs**” for the unfiltered ranked table, “**Pathway Enrichment**” for the full enrichment table.

Report generated by GEO Melanoma project — 2026-06-05 16:26:36.167198 Pipeline: fastp + subjunc + featureCounts + DESeq2 + clusterProfiler + ReactomePA + decoupleR/DoRothEA Deliverables: PDF report + Excel workbook (results_workbook.xlsx) + MultiQC HTML For questions contact: hello@omicsdesk.com